Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

ctrl + ai

Selected Domain

Domain 1: The Aiversecity Future-Ready Workforce Domain

Selected SDG Track (1–17)

4 - Quality Education; supporting SDG 8

Project / Solution Name

SkillHalfLife - with AI Trend Signal evidence layer

Name (Leader)

Ahmad Irfan Bin Ishak

Email & Phone (Leader)

ahmadirfan2010@gmail.com, +60176648750

University Affiliation(s)

University Teknologi Malaysia (UTM)

Number of Team Members

5

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Students, fresh graduates and early-career workers choosing their next learning investment face three disconnected questions: which skills they already have, which changes in AI and hiring deserve attention, and what affordable learning to do next. Generic course lists and unexplained job rankings leave users unable to distinguish evidence from speculation.

The World Economic Forum's Future of Jobs Report 2025 reports that surveyed employers expect 39% of workers' existing skill sets to change or become outdated during 2025-2030; 63% identify skill gaps as a major barrier to transformation [1]. These are global employer expectations, not measured Malaysian displacement rates or forecasts for individual users.

Our initial focus is Malaysian university students and early-career technology workers making a near-term reskilling decision, with career advisers as supporting users. The economic challenge is spending limited time and money on learning that may not address a person's actual gaps. The social challenge is unequal access to understandable guidance and affordable training.

SkillHalfLife connects a reviewed skills profile, dated public evidence, explainable role comparisons and an ordered learning plan. Users retain control over their choices and see uncertainty explicitly. The intended contribution is better access to employment-relevant learning under SDG 4, supporting SDG 8; we do not claim guaranteed employment, higher wages or measured carbon savings.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SkillHalfLife is a bilingual English/Bahasa Malaysia career-exploration application supported by the AI Trend Signal evidence layer. It turns a reviewed skills profile into ranked technology-role options and a practical reskilling pathway.

First, a user loads a synthetic CV or completes a questionnaire, reviews extracted skill mentions and confirms self-assessed levels. The demonstration excludes real CVs and personal data. Second, the Future Skills Radar presents dated Remotive remote-job signals and arXiv AI-research metadata. OpenDOSM labour-force indicators provide separate Malaysian context, not skill-demand estimates.

Third, an explainable recommendation engine combines skills (70%), interests (20%) and education/experience (10%). SDG preference points and bounded evidence-based scenario adjustments are shown separately. Moderate and fast scenarios expose assumptions rather than promise a single future. Fourth, the pathway orders skill gaps by prerequisites, links verified public learning resources, estimates practice time and records browser-local progress. Each recommended role and skill explains its SDG contribution.

The current prototype uses transparent rules, keyword extraction and scenario modelling; it does not use a trained predictive model or GenAI API. For Phase 2, we propose an optional retrieval-grounded GenAI explanation layer restricted to approved evidence, with citations and deterministic fallback. It must not invent jobs, courses, proficiency levels or statistics. Provider/model selection and sponsor eligibility remain to be confirmed.

The implementation uses workshop-relevant API integration, relational database design and cloud-deployment skills through React/TypeScript, Express and PostgreSQL. The outcome is a source-linked plan users can question and revise, not an automated hiring decision.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Interface: React, TypeScript, Vite, Tailwind CSS, Recharts, TanStack Query and Wouter; English/Bahasa Malaysia and responsive navigation.

Service: Node.js, Express, Zod validation and OpenAPI/Orval-generated contracts. PDF/DOCX intake uses bounded, temporary parsing workers; users must review extracted mentions.

Storage: PostgreSQL with Drizzle ORM stores normalized public evidence, snapshots and refresh history. Reviewed demo profiles are tab-scoped; learning progress is browser-local. CV files are not retained.

Evidence: Remotive remote-job API, arXiv research API and OpenDOSM labour-force API. O*NET supplies related occupational references; approved learning-provider URLs support pathways.

Intelligence: deterministic matching, prerequisite ordering and disclosed scenario assumptions. No GenAI model/API is currently integrated.

Hosting: existing Replit development workflows; separate frontend and shared API. A public Phase 2 release is planned, not certified here. Appendix A shows the system flow.

Official SDC sponsor GenAI API/cloud: not verified in the supplied project. Team confirmation is required; Replit hosting must not be described as an official sponsor service without verification.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our design-thinking process starts with a focused decision: a learner with limited study time needs to choose an achievable next skill, not receive an unexplained career prediction. This is a design hypothesis; formal user interviews and outcome validation have not yet been completed.

We defined a four-step journey: profile, evidence, role comparison and pathway. The implemented prototype supports synthetic examples, manual intake, explicit profile review, English/Bahasa Malaysia, mobile layouts, source dates and clear reset controls. Deterministic explanations and separate observed-versus-simulated views make uncertainty inspectable.

For Phase 2, consenting testers will use only synthetic profiles in a counterbalanced comparison against manual source browsing. We will measure completion, time and evidence comprehension, then revise confusing steps. No names, real CVs or demographic attributes are needed.

Deployment readiness will be assessed through reproducible builds, parser/input checks, source-outage tests, accessibility checks and a public demonstration linked to an accessible repository. Server-side refresh limits and stored-data fallback protect provider services. Automated ingestion and the optional grounded GenAI layer remain planned, not completed.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

PROJECT IMPACT BASIS

Primary track: SDG 4 - Quality Education; supporting SDG 8.
The framework below combines a published global problem baseline with an explicitly estimated pilot baseline. No pilot result is claimed.

References and the measurement protocol are attached as appendices.
The prototype does not establish emissions savings, job placement, wage uplift or achievement of UN national-level indicators.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: SDG 4, Target 4.4 - increase relevant technical and vocational skills for employment, decent jobs and entrepreneurship [2]. Supporting: Target 4.3 - equal access to affordable, quality technical, vocational and tertiary education [2]; Target 8.5 - productive employment and decent work [3]. Contribution: show skill gaps and affordable, ordered learning choices. Employment effects are indirect and not established by this prototype. Links to all 17 goals support exploration; they do not prove impact.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Published context: employers surveyed by WEF expect 39% of existing skills to change/become outdated by 2030; 63% identify skills gaps as a major barrier [1]. These are global survey expectations. Pilot baseline assumption: 20 minutes to produce a source-linked learning plan through manual browsing. This is a planning estimate, not a measured result. Before testing the app, measure actual median manual-task time using synthetic profiles. Replace the 20-minute assumption with that result. No verified employment, learning-gain or environmental baseline exists yet.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Primary KPI: median time (minutes) to complete a valid, source-linked plan from an assigned synthetic profile. Valid plan = a chosen role, two explained skill gaps, ordered learning steps, working resource URLs, estimated effort and a concrete SDG link. Improvement (%) = 100 x (manual median - app median) / manual median. Guardrails: task completion rate, source traceability, evidence-versus-scenario comprehension, broken links and unsupported claims. See Appendix B for measurement.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Phase 2 target: at least 30% lower median planning time than the measured manual baseline, without reducing valid-plan completion. Illustration only: 20 minutes manually to 14 minutes with the app. This is a target, not an achieved result. Proposed pilot: 12 consenting testers, paired equivalent synthetic tasks and counterbalanced order. Seek at least 80% valid-plan completion and 100% source-linked role/resource recommendations.

Report raw results, failures and uncertainty. These are prototype usability indicators, not proof of SDG target attainment, employment uplift or carbon reduction.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React, Express, UI / query libraries	Code / frameworks; full inventory in Appendix D	MIT / ISC / Unlicense; see D
TypeScript, Drizzle; parser libraries	Code / build / ORM / parsing	Apache-2.0, BSD-2-Clause, MIT; see D
OpenDOSM; O*NET database	Public datasets / occupational references	CC BY 4.0; attribution required
Remotive; arXiv API	Job feed; research metadata. No pretrained AI model is integrated.	Provider terms; arXiv metadata CC0; abstract rights vary
Learning providers	Linked learning materials; not republished course content	Provider-specific; see Appendix C

Yes: third-party dependencies and public evidence are declared. Preserve licence notices; verify exact locked versions and provider terms before redistribution.

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

25/10/2026

Appendix A | Architecture & delivery

Scope: SkillHalfLife is the user-facing Domain 1 product; AI Trend Signal is its supporting evidence service, not a second unrelated proposal.

System block diagram

SYNTHETIC PROFILE	PUBLIC EVIDENCE
Sample PDF/DOCX or manual questionnaire Review + explicit confirmation	Remotive | arXiv | OpenDOSM Timeouts, validation, provider cooldowns
Reviewed tab-scoped profile	Normalized records + source dates PostgreSQL / Drizzle + refresh history
EXPLAINABLE ENGINE	**EVIDENCE CONTROLS**
Skill / interest / education matching SDG preferences + bounded scenarios	Equal-window comparisons Insufficient-data and stale-source states
USER OUTPUT	**PLANNED GENAI EXTENSION**
Ranked roles -> gaps -> ordered pathway Resource links + time + SDG mechanism Human chooses; local progress only	Approved evidence retrieval -> cited explanation Never authoritative for scoring Model/provider TBD; rule-based fallback

Data flow: the browser calls the Express API. A manual refresh fetches provider records, normalizes/upserts them and saves source status. Catalog/evaluation endpoints combine stored public evidence with the confirmed synthetic profile. Browser polling re-reads stored data; scheduled external ingestion is not implemented.

Demonstration and access

Suggested demo: load a synthetic profile; confirm levels; inspect evidence dates; compare role scores; change moderate/fast scenario; open a learning resource and explain its SDG link. Controlled synthetic evidence fixtures can demonstrate ranking sensitivity, but must be labelled test fixtures, never live market data.

Live improved-app URL: [TEAM TO PROVIDE AFTER PUBLISHING]. GitHub repository: [TEAM TO PROVIDE; public or accessible to judges]. The previously supplied original-site link is not represented as the improved deployment.

Appendix B | Evaluation & Phase 2 plan

Measurement protocol

Proposed sample: 12 consenting testers using synthetic personas only. Each performs two equivalent role-and-learning planning tasks: one using manual browsing of the approved sources and one using SkillHalfLife. Counterbalance order (six manual first, six app first) and assign matched but different personas to reduce recall effects. Do not collect real CVs, names or demographic attributes.

Start timing when the assigned profile is revealed; stop when the plan meets the predefined valid-plan rubric. Use a 30-minute task cap; report non-completions separately and do not exclude slow or failed tasks to improve the KPI. Report paired changes, medians, ranges, completion counts and order effects; a small convenience pilot does not establish population-level effectiveness.

Valid-plan rubric: chosen role; two profile-specific skill gaps; prerequisite-aware sequence; source-linked learning materials; estimated practice time; at least one concrete role/skill-to-SDG explanation. Source comprehension: the tester must distinguish global remote postings, Malaysian national labour statistics, AI research activity and hypothetical scenario assumptions.

Primary target: >=30% lower median planning time, with app valid-plan completion no lower than manual completion. Additional gates: >=80% app completion; every recommended role and learning resource linked to evidence; no invented courses/statistics; source outages clearly labelled. The 20-to-14-minute example is an assumption, not a reported result.

Technical gates and highest risks

Evidence coverage is the hardest risk: sparse or biased global remote data can mislead Malaysian learners. Keep streams separate, display coverage and dates, suppress weak signals and avoid national vacancy claims. Preserve stored evidence during outages. A market adjustment requires at least five mentions in each equal 30-day window; unavailable evidence contributes zero.

Matching remains uncalibrated. Validate synthetic edge cases, explain missing information and test whether SDG preferences or small input changes create unstable rankings. No demographic attributes enter matching; that alone is not proof of fairness. Moderate/fast half-lives, skill thresholds and practice budgets are illustrative, not employment forecasts.

Ordered delivery milestones

1. Establish the manual baseline, role/resource citation audit and reproducible tests. 2. Refine the mobile/profile-to-pathway flow from synthetic-task observations. 3. Add scheduled ingestion only within provider limits and with failure visibility. 4. If a GenAI provider is selected, restrict it to approved retrieved evidence, treat source text as untrusted input and reject unsupported output. 5. Publish a separate demo, verify it, and supply the repository and limitations to judges.

Existing engineering evidence: 22 model/parser tests and prior browser checks support prototype feasibility. These are not human pilot results, measured learning gains, employment outcomes or proof of production reliability.

Appendix C | Citations & resource provenance

[1] World Economic Forum (2025). The Future of Jobs Report 2025, digest. Global employer survey; 39% skills instability and 63% skills-gap barrier.
https://www.weforum.org/publications/the-future-of-jobs-report-2025/digest/

[2] United Nations. Goal 4: Targets 4.3 and 4.4. Official target wording; project indicators are not official SDG indicators.
https://sdgs.un.org/goals/goal4

[3] United Nations. Goal 8: Target 8.5. Employment contribution is aspirational, not a measured causal outcome.
https://sdgs.un.org/goals/goal8

[4] DOSM. Monthly Principal Labour Force Statistics. Malaysia national survey; CC BY 4.0. Not vacancies or skill demand.
https://open.dosm.gov.my/data-catalogue/lfs_month

[5] Remotive. Public remote-jobs API and usage conditions. Returned global remote feed; delayed by at least 24 hours; retain attribution.
https://remotive.com/remote-jobs/api

[6] arXiv. API terms of use. Metadata is available under CC0; abstracts/full text have separate rights and must not be assumed CC0.
https://info.arxiv.org/help/api/tou.html

[7] O*NET Resource Center. Database and licensing. Related US occupations, not validation of model thresholds or Malaysian outcomes.
https://www.onetcenter.org/license_db.html

[8] Aiversecity / SDC 2026. Domain 1 problem statement and marking rubric. Supplied competition brief; synthetic/public data only and accessible GitHub repository required.
User-supplied Domain_1_1792852409164.pdf

Access check: the web references [1]-[7] were reviewed during submission preparation. Learning-provider links below were checked for the prototype on 24 October 2026. Provider pricing, availability and licence terms may change. Free learning access does not mean a free certificate, exam, cloud runtime or course-content redistribution right.

Pathway learning-resource registry

Programming: Harvard CS50
https://cs50.harvard.edu/x/

Cloud: Microsoft Learn
https://learn.microsoft.com/en-us/training/paths/microsoft-azure-fundamentals-describe-cloud-concepts/

Cybersecurity: PortSwigger Web Security Academy
https://portswigger.net/web-security

System design: System Design Primer (community)
https://github.com/donnemartin/system-design-primer

AI literacy: University of Helsinki / MinnaLearn
https://www.elementsofai.com/

Communication: Google Technical Writing
https://developers.google.com/tech-writing

Databases: SQLBolt
https://sqlbolt.com/

Networking: Cisco Networking Basics
https://www.netacad.com/courses/networking-basics

Machine learning: Google Machine Learning Crash Course
https://developers.google.com/machine-learning/crash-course

Data pipelines: Apache Airflow tutorials
https://airflow.apache.org/docs/apache-airflow/stable/tutorial/index.html

arXiv acknowledgement: Thank you to arXiv for use of its open access interoperability.

Appendix D | Third-party declarations

The following direct runtime/development dependencies were identified from the supplied product packages and matching installed package metadata. SPDX labels describe software licensing, not permission to reuse trademarks or course content. Exact locked versions, transitive dependencies, bundled assets and required NOTICE files must be audited before redistribution; this inventory is not legal clearance.

Dependency	Declared licence
@hookform/resolvers	MIT
@radix-ui/react-accordion	MIT
@radix-ui/react-alert-dialog	MIT
@radix-ui/react-aspect-ratio	MIT
@radix-ui/react-avatar	MIT
@radix-ui/react-checkbox	MIT
@radix-ui/react-collapsible	MIT
@radix-ui/react-context-menu	MIT
@radix-ui/react-dialog	MIT
@radix-ui/react-dropdown-menu	MIT
@radix-ui/react-hover-card	MIT
@radix-ui/react-label	MIT
@radix-ui/react-menubar	MIT
@radix-ui/react-navigation-menu	MIT
@radix-ui/react-popover	MIT
@radix-ui/react-progress	MIT
@radix-ui/react-radio-group	MIT
@radix-ui/react-scroll-area	MIT
@radix-ui/react-select	MIT
@radix-ui/react-separator	MIT
@radix-ui/react-slider	MIT
@radix-ui/react-slot	MIT
@radix-ui/react-switch	MIT
@radix-ui/react-tabs	MIT
@radix-ui/react-toast	MIT
@radix-ui/react-toggle	MIT
@radix-ui/react-toggle-group	MIT
@radix-ui/react-tooltip	MIT
@replit/vite-plugin-cartographer	Check upstream licence
@replit/vite-plugin-dev-banner	Check upstream licence
@replit/vite-plugin-runtime-error-modal	Check upstream licence

Dependency	Declared licence
@tailwindcss/typography	MIT
@tailwindcss/vite	MIT
@tanstack/react-query	MIT
@tanstack/react-table	MIT
@types/cookie-parser	MIT
@types/cors	MIT
@types/express	MIT
@types/multer	MIT
@types/node	MIT
@types/papaparse	MIT
@types/pg	MIT
@types/react	MIT
@types/react-csv	MIT
@types/react-dom	MIT
@types/yauzl	MIT
@vitejs/plugin-react	MIT
class-variance-authority	Apache-2.0
clsx	MIT
cmdk	MIT
cookie-parser	MIT
cors	MIT
d3-scale-chromatic	ISC
date-fns	MIT
drizzle-kit	MIT
drizzle-orm	Apache-2.0
drizzle-zod	Apache-2.0
embla-carousel-react	MIT
esbuild	MIT
esbuild-plugin-pino	MIT
express	MIT
express-rate-limit	MIT
fast-xml-parser	MIT
framer-motion	MIT
input-otp	MIT
lucide-react	ISC
mammoth	BSD-2-Clause
multer	MIT

Dependency	Declared licence
next-themes	MIT
papaparse	MIT
pdf-parse	Apache-2.0
pg	MIT
pino	MIT
pino-http	MIT
pino-pretty	MIT
react	MIT
react-csv	MIT
react-day-picker	MIT
react-dom	MIT
react-hook-form	MIT
react-icons	MIT
react-resizable-panels	MIT
recharts	MIT
sonner	MIT
tailwind-merge	MIT
tailwindcss	MIT
thread-stream	MIT
tw-animate-css	MIT
vaul	MIT
vite	MIT
wouter	Unlicense
yauzl	MIT
zod	MIT

Data and content: OpenDOSM and the O*NET database use CC BY 4.0 with attribution and applicable terms. Remotive use follows its API conditions, source links and rate limits. arXiv API metadata is CC0; abstracts and papers are not presumed CC0. Third-party course pages are linked, not copied. The current prototype uses no pretrained AI model, model weights or external GenAI API. A future model/service requires its own provider and terms disclosure.